Fundamentals of geospatial science and technology

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Topic: Assignment 2 – *Map your own world*

SPATIAL DISTRIBUTION AND OBSERVED CHARACTERISTICS OF SCHOOLS IN BAVDHAN

GIS Assignment 2: Map Your Own World | Field survey conducted 11 October 2026

Research Question

How are the surveyed schools distributed in and around Bavdhan (both Government and private), and what do their observed entrance access, boundary condition, and playground availability indicate about their visible characteristics?

Schools are important neighbourhood facilities, and their locations and visible characteristics can help describe the local pattern of educational infrastructure. This mini-GIS project records school locations in and around Bavdhan and maps the schools together with selected field-observed attributes. The survey focuses on **school management (Government or private), educational level, entrance accessibility, visible boundary condition, and playground availability**.

Methodology

Primary data were collected using an EpiCollect field form on 10 and 11 October 2026. The exported dataset contains 21 school records. Each record includes the school name, management type, school level, observed entrance access, visible boundary condition, playground status, latitude and longitude, and a photo filename. The coordinates were captured in decimal degrees; the recorded coordinate range is approximately 18.509077–18.524387° N and 73.761719–73.782211° E. The point locations were imported into ArcGIS Pro and displayed over a basemap to examine their distribution in relation to neighbourhoods and roads.

The attributes were selected to combine nominal categories (management and playground availability), ordinal observations (entrance access and boundary condition), and geographic coordinates.

Findings and Spatial Pattern

The survey recorded 21 schools. The management field contains **18 private schools (85.7%)** and **3 government schools (14.3%)**. Therefore, private schools form the majority of the surveyed sample. This describes only the schools recorded within the selected survey area; it should not be interpreted as a complete census of all schools in the wider region.

For **entrance access, 17 schools were recorded as "Good" (81.0%) and 4 as "Moderate" (19.0%)**. No record was coded "Poor". (These ratings reflect the field observer's visible assessment of the entrance and approach, rather than a formal accessibility audit).

The visible **boundary condition** was recorded as **"Good" for 13 schools (61.9%)** and **"Fair" for 8 schools (38.1%)**. No "Poor" rating recorded. The observations suggest that most surveyed school boundaries were assessed as good, although the ratings are qualitative and based on visible conditions.

["Fair" boundary was marked on the basis of how safe the boundary of the school is, in these cases many of the primary schools were located right along the busy roads with makes their boundaries and entrances unsafe for kids.]

Playgrounds were recorded as **available at 10 schools (47.6%), not available at 10 schools (47.6%),** and **"Not observable" at 1 school (4.8%).** The nearly equal counts of available and not-available playgrounds show variation in this visible facility among the surveyed schools.

The concept of an open playground for kids (in preprimary schools) has vanished from the urban areas because of the area constraint in congested city spaces.

Note: "Playground" that I've considered In my report is the "open-air" ground with larger area and ground with mud or grass. Thus, 8 out of 10 'not available playground' schools actually had an indoor play area for kids, visibly complying with the MSBSHSE (State Board) norms for children's play areas in urban preprimary schools [protective flooring, such as rubberized mats, artificial turf, or thick sand beds, to avoid hard concrete falls].

The mapped points appear concentrated along the built-up neighbourhood corridor around Patil Nagar, Siddharth Nagar and Bavdhan, with a smaller number of points extending toward the western and outer parts of the survey area. This indicates a clustered rather than evenly spaced pattern in the collected sample. The pattern reflects the location of built-up residential areas and the route followed during field collection.

Private vs Government schools:

Private schools clearly dominate (18/21). The less occurrence of government schools (3) fits into the RTE Act criteria of – Primary School (classes 1 to 5) must be established within a walking distance of 1 kilometer of any habitation; but the visible condition of boundaries, entrance and area occupied by these schools is not as compatible as their private counterparts.

The distance from the main roads, and accessibility of government schools is moderate to good but the number of these schools and other facilities should be improved to meet the requirements of concentrating population and continuous urban sprawl.

Only 8 schools are providing secondary or higher secondary classes education out of 21 surveyed, this shows the abundance in the preprimary education facilities in Bavdhan. And out of these 8 only 3 schools are facilitating higher secondary education (11 & 12).

Map

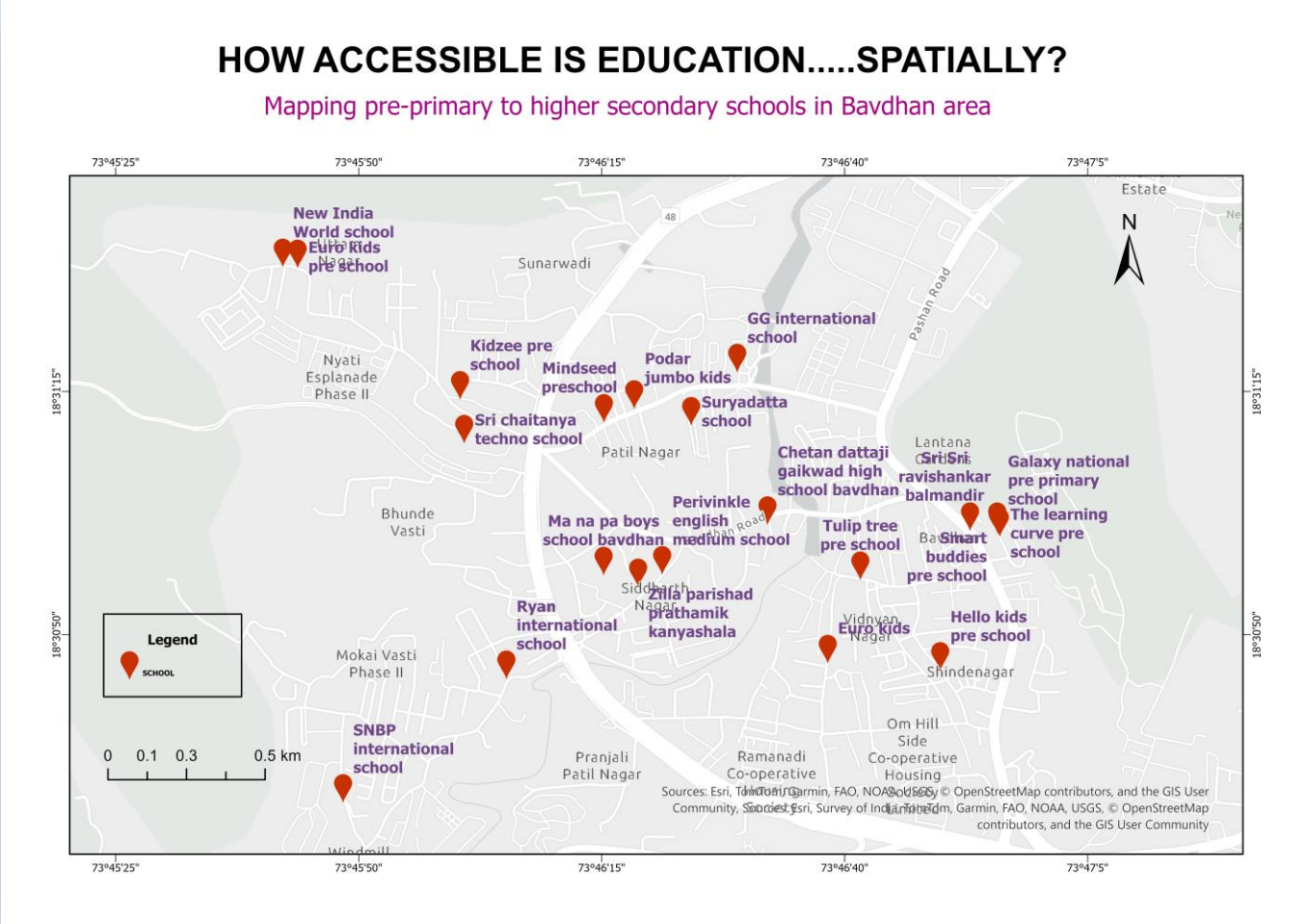


Figure 1. Surveyed school locations in and around Bavdhan

Conclusion

The field survey successfully documented 21 school locations and collected comparable observations on management, entrance access, boundary condition and playground availability. The sample is dominated by private schools, most recorded entrances were rated good, and playground availability varied across the surveyed schools. The map suggests a concentration of recorded schools around the main built-up neighbourhood corridor rather than a uniform distribution.

Personal Reflection

The importance of primary education and easy availability and accessibility of data encouraged me to choose this topic for the assignment. I learnt to build EpiCollect form, adding the schema and questions required by the research question to be answered, and visualizing the collected data by importing it into arcgis pro and drawing out the collected points on the map. The data collection experience was exciting overall, although at one of the schools I faced difficulty to take photograph of the school due to security reasons.

One pattern on the map that surprised me was the cluster of preprimary schools even in such a small area. I even saw 3-4 preprimary schools within distance of less than 500 meters. This can be inferred as the new pattern emerging in urban spaces for the ease of both working parents and safety concerns of the children and saving the time in travelling distance among others.

Attribute Schema / Data Dictionary Table

Field name	Data type	Scale	Allowed values / units	Why it matters
Entry number	Number	Ratio	1, 2, 3	Provides a unique identification code for each school.
Management	Text	Nominal	Government / Private / Other	Helps compare the spatial distribution of different types of schools.
School_Level	Text	Nominal	Preschool / Primary / Secondary / Higher Secondary / Combined	Identifies the educational levels served by each school.
Entrance_Access	Text	Ordinal	Good / Moderate / Poor	Assesses the observed accessibility of the school entrance and approach.
Boundary_Condition	Text	Ordinal	Good / Fair / Poor	Records the visible condition of the school's boundary wall or fence.
Playground	Text	Ordinal	Available / Not available / Not observable	Identifies whether a playground is visibly available at the school.
Latitude / Longitude	Number		decimal degrees	Location of the point
Photo	Link		file name / photo	Field evidence